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COMPUTER

CHAPTER 21

Data Encryption and Cryptography

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COMPUTER · CHAPTER 21

Data Encryption and Cryptography

ਡਾਟਾ ਇਨਕ੍ਰਿਪਸ਼ਨ ਅਤੇ ਕ੍ਰਿਪਟੋਗ੍ਰਾਫੀ

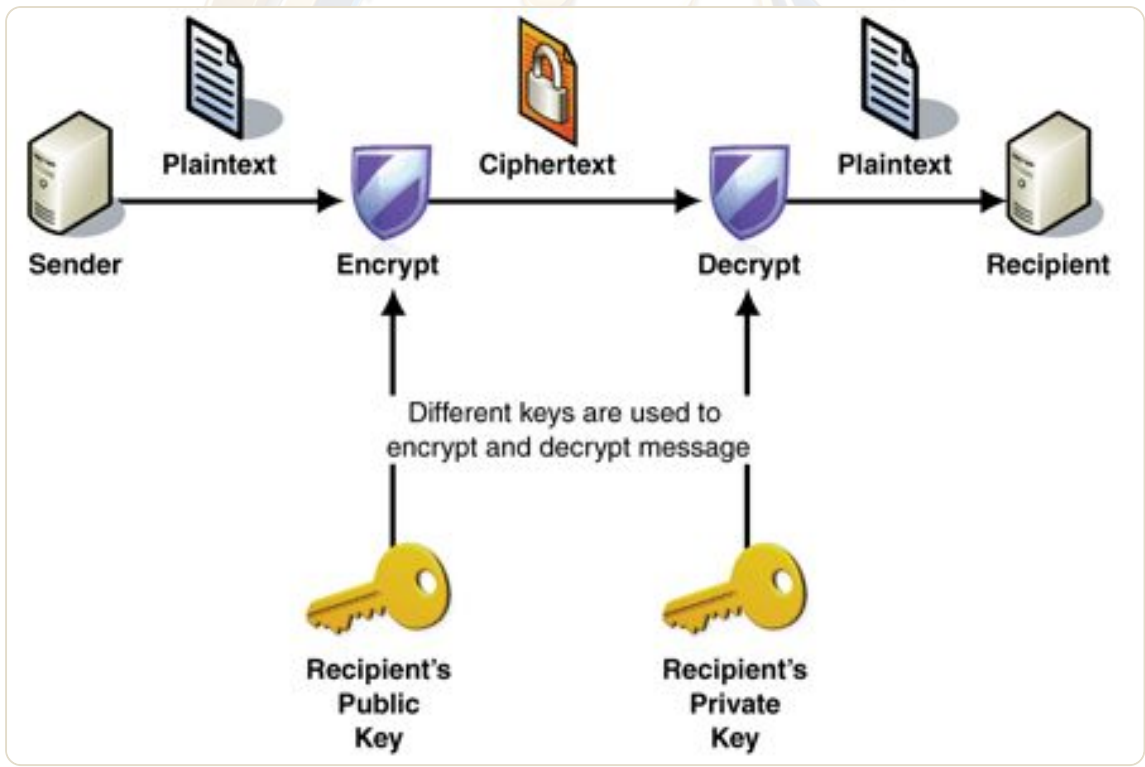
IN THIS CHAPTER

- Cryptography
- Basic Terms In Cryptography
- Types Of Encryption
- Symmetric Encryption
- Symmetric Encryption Algorithms
- Asymmetric Encryption

CRYPTOGRAPHY

Cryptography is the science and art of protecting information by converting it into an unreadable format so that unauthorized persons cannot access it.

Example



Original Message: **HELLO**

Encrypted Message: **4873r@#$**

Only authorized users can convert it back to the original message.

BASIC TERMS IN CRYPTOGRAPHY

Sender :The person or system that sends the information.

Recipient (Receiver): The person or system that receives the information.

Plaintext:The original readable information before encryption.

Example: HELLO

Ciphertext:The encrypted unreadable form of plaintext.

Example: 4873r@#$

Encryption: The process of converting plaintext into ciphertext using an encryption algorithm.

Example: HELLO → Encryption → 4873r@#$

Decryption: The process of converting ciphertext back into plaintext using a decryption algorithm.

Example:

4873r@#$ → Decryption → HELLO

TYPES OF ENCRYPTION

There are two main types of encryption:

1. Symmetric Encryption

2. Asymmetric Encryption

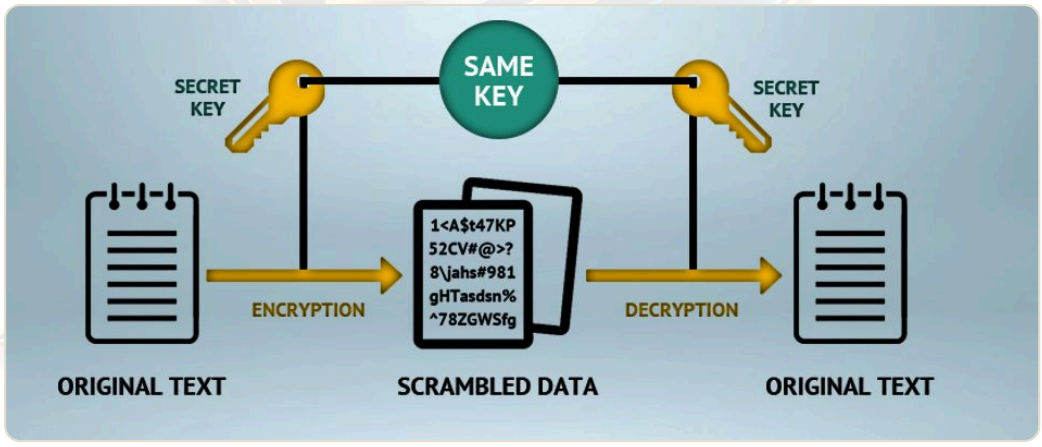
SYMMETRIC ENCRYPTION

In Symmetric Encryption, the **same key** is used for both encryption and decryption.

The sender and receiver must know and share the same secret key beforehand.

Also Known As

- Secret Key Encryption
- Private Key Encryption

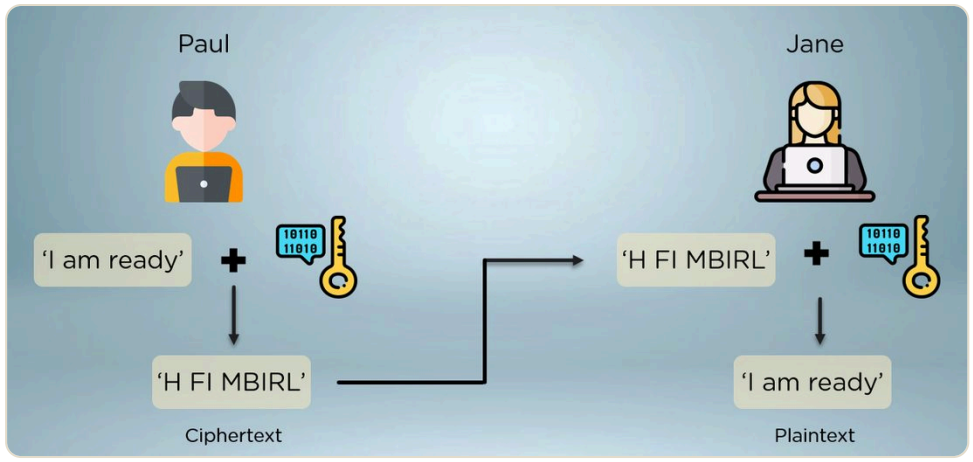


Example

Key = 12345

Sender encrypts the message using key 12345.

Receiver decrypts the message using the same key 12345.



Types of Symmetric Ciphers

A. Stream Cipher

Encrypts data one bit or one byte at a time.

B. Block Cipher

Encrypts data in fixed-size blocks.

Example

A 128-bit block cipher divides plaintext into blocks of 128 bits and encrypts each block separately.

SYMMETRIC ENCRYPTION ALGORITHMS

1. DES (Data Encryption Standard)

Features

- Developed by IBM
- Block size: 64 bits
- Key size: 56 bits
- Block Cipher

Limitation

Not secure today because the key size is too small.

2. Double DES

DES encryption is applied twice to increase security.

3. Triple DES (3DES)

DES is performed three times:

Encrypt → Decrypt → Encrypt

4. AES (Advanced Encryption Standard)

Features

- Current international encryption standard
- Considered the "Gold Standard" of encryption
- Developed by Joan Daemen and Vincent Rijmen; adopted as a standard by NIST
- Key Sizes:
  - 128-bit

- 192-bit
- 256-bit

Uses

- Banking
- Government systems
- Wi-Fi security
- VPNs
- Cloud storage

5. Blowfish

Features

- Fast symmetric encryption algorithm
- Free to use
- Key size: 32 to 448 bits

6. Twofish

Features

- Successor to Blowfish
- Very fast
- Highly secure
- Free to use

ASYMMETRIC ENCRYPTION

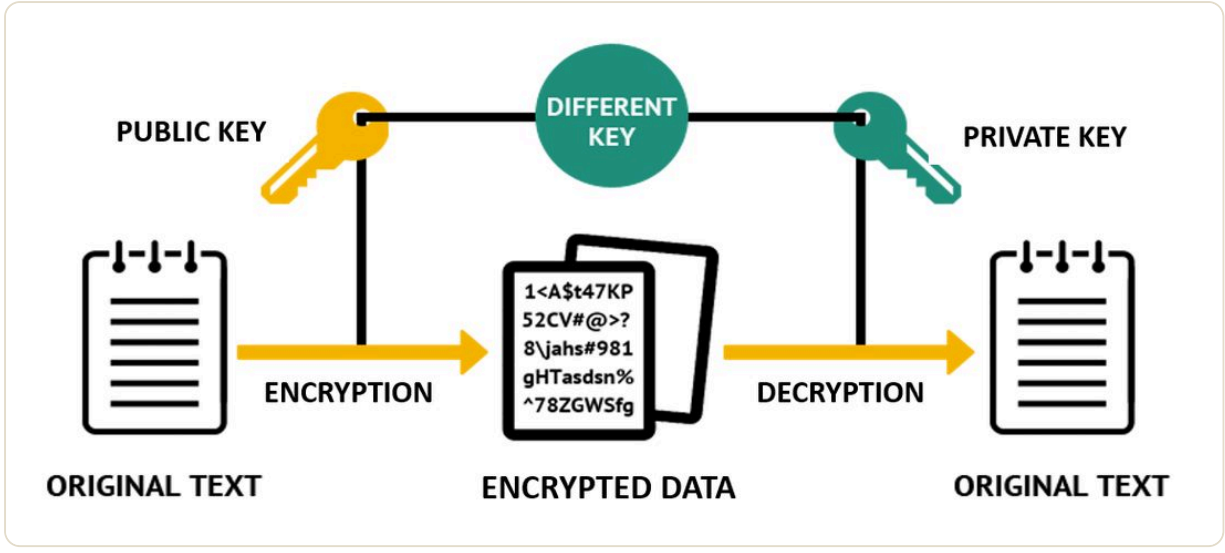
Asymmetric Encryption uses **two different keys**:

1. Public Key

2. Private Key

Also Known As

- Public Key Cryptography
- Public Key Encryption



WORKING

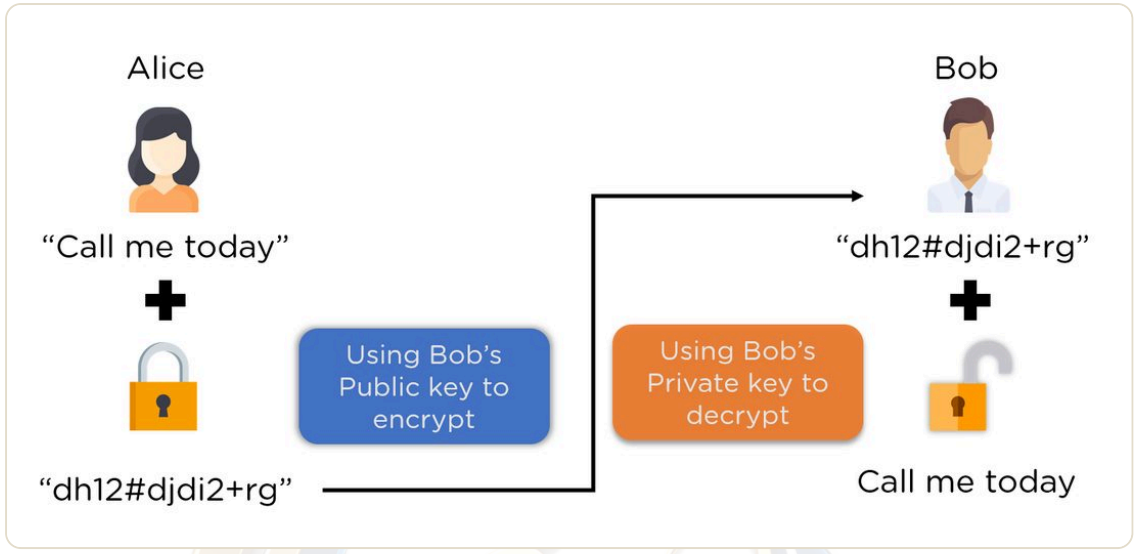
Public Key → Encryption

Private Key → Decryption

Both keys belong to the receiver.

Example

Suppose Alice wants to send a message to Bob.



Bob owns:

- Public Key
- Private Key

Alice encrypts the message using Bob's Public Key.

Bob decrypts the message using his Private Key.

Important Note

Rule 1

Message encrypted with Public Key

↓

Can only be decrypted with Private Key

Rule 2

Message encrypted with Private Key

↓

Can only be decrypted with Public Key

(Used in Digital Signatures)

ASYMMETRIC ENCRYPTION ALGORITHMS

1. RSA (Rivest–Shamir–Adleman)

Features

- Most popular asymmetric algorithm
- Used in:
  - VPNs
  - Email encryption
  - SSL/TLS
  - Digital Signatures

Key Size

- 1024-bit
- 2048-bit
- 4096-bit (modern systems)

2. Diffie-Hellman Key Exchange

Developed By

- Whitfield Diffie
- Martin Hellman

Purpose

Used to securely exchange cryptographic keys over an insecure network.

Exam Point

Diffie-Hellman is mainly used for key exchange, not direct encryption.

3. DSA (Digital Signature Algorithm)

Used specifically for digital signatures and authentication.

4. ElGamal Algorithm

Public-key cryptographic algorithm used for:

- Encryption
- Digital Signatures

5. PKI (Public Key Infrastructure)

PKI is a framework that manages digital certificates and public keys.

6. DES(data encryption standard)

Data Encryption Standard (DES) is a block cipher with a 56-bit key length that has played a significant role in data security.

DES is a block cipher and encrypts data in blocks of size of **64 bits** each and uses a 56 bit key.

It is founded by IBM .

it uses 56 bit key size

HASHING

Hashing is a process that converts data into a fixed-length value called a Hash Value or Hash Code.

Unlike encryption:

No key is used

One-way process

Original data cannot be recovered from the hash

Example

Message: HELLO

Hash: EB61EEAD90E3B899C6BCBE27AC581660

Uses of Hashing

- Password Storage
- Data Integrity Verification
- Digital Signatures
- Blockchain

MESSAGE DIGEST

A Message Digest is a fixed-size hash value generated from a message.

Purpose

To verify data integrity.

Exam Point

Message Digest Algorithm (MD5)

- Produces 128-bit hash value
- Widely used historically
- No longer considered secure for cryptographic security

SHA (SECURE HASH ALGORITHM)

Developed By NIST (National Institute of Standards and Technology), USA

SHA Family

SHA-0

- 160-bit hash
- Obsolete

SHA-1

- 160-bit hash
- Vulnerable to attacks
- Not recommended

SHA-2

Most widely used secure hash family.

Variants:

- SHA-224
- SHA-256

- SHA-384
- SHA-512

Hash sizes:

224 to 512 bits

Exam Point

SHA-256 is commonly used in modern security systems.

SHA-3

Latest SHA family.

Provides additional security against future attacks.

PUBLIC KEY CERTIFICATE

A Public Key Certificate (Digital Certificate) is an electronic document that proves ownership of a public key.

KEY DIFFERENCES

SYMMETRIC VS ASYMMETRIC ENCRYPTION

Feature	Symmetric Encryption	Asymmetric Encryption
Keys Used	One Key	Two Keys
Speed	Faster	Slower
Security	Lower	Higher
Key Sharing	Required	Not Required
Examples	AES, DES, Blowfish	RSA, DSA, ElGamal
Use	Large Data Encryption	Key Exchange, Digital Signatures

MOST IMPORTANT EXAM FACTS

1. AES is the current encryption standard.

2. DES uses a 56-bit key and is now insecure.

3. Triple DES performs DES three times.

4. RSA is the most widely used public-key algorithm.

5. Diffie-Hellman is used for key exchange.

6. DSA is used for digital signatures.

7. MD5 generates a 128-bit hash.

8. SHA-1 produces a 160-bit hash.

9. SHA-256 belongs to SHA-2 family.

10. Hashing is one-way and cannot be decrypted.

11. Public Key encrypts, Private Key decrypts.

12. Digital Certificates are issued by Certificate Authorities (CA).

13. PKI manages public keys and digital certificates.

QUICK REVISION NOTES

- Cryptography = Protection of information.
- Plaintext = Readable data.
- Ciphertext = Encrypted data.
- Encryption = Plaintext → Ciphertext.
- Decryption = Ciphertext → Plaintext.
- Symmetric Encryption = Same key.
- Asymmetric Encryption = Public Key + Private Key.
- AES = Most secure and widely used symmetric algorithm.
- RSA = Most popular asymmetric algorithm.
- Diffie-Hellman = Key exchange.
- DSA = Digital Signature.
- Hashing = One-way process.
- MD5 = 128-bit hash.
- SHA-1 = 160-bit hash.
- SHA-256 = Most commonly used SHA version.
- PKI = Manages certificates and public keys.
- Digital Certificate = Proof of ownership of a public key.